

EFFICACY OF OKRA MUCILAGE COATINGS ON STORAGE-LIFE EXTENSION OF FRESHLY HARVESTED TOMATOES

PROPONENT:

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CATEGORY: Life Science

THEME: Food Safety

CHAPTER 1: INTRODUCTION

Tomato post-harvest losses significantly impact the economy, environment, and livelihoods of people. Studies show that **edible coatings** made with aloe vera gel help preserve and extend their shelf lives. Given that okra contains polysaccharides, a barrier to moisture and O_2 , this **objective of this study** to determine whether okra mucilage can effectively preserve tomatoes under cold storage conditions.

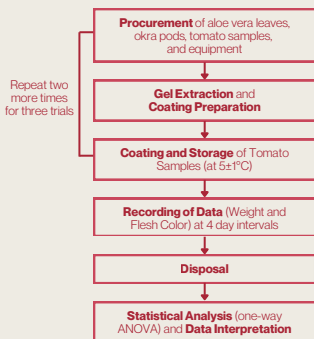
CHAPTER 2: LITERATURE REVIEW

Edible Polysaccharide Coatings: Polysaccharides in aloe vera (*aloe barbadensis miller*) gel serve as a barrier to O_2 (dioxide) and moisture, which slows down the respiration rate of fruit and maintains its quality (Rehman et al., 2020; Maan et al., 2018). It is also one of the main bioactive components of okra (*A. esculentus*).

Efficacy of Aloe Vera Coatings: Research by Kator et al. (2018) show that tomatoes treated with gel coated at 100% concentration yielded the highest marketability and firmness ratings, best weight, and lowest decay percentage. A study by Roy and Karmakar (2019) showed similar results wherein the coated tomatoes had longer shelf lives and lower decay percentage.

CHAPTER 3: METHODOLOGY

Figure 1. Flowchart of Methodology



CHAPTER 4: RESULTS

Figure 2. Weight Loss of Tomatoes

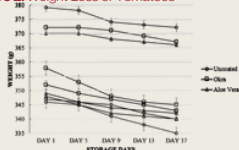


Figure 4. Mean Lightness (L^*) Values of Tomatoes

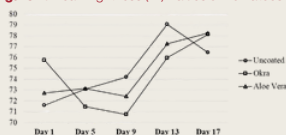


Figure 5. Mean Red/Green (a^*) Values of Tomatoes

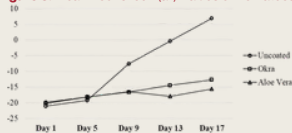
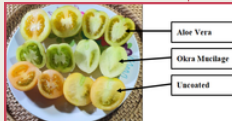


Figure 6. Photo Documentation of Tomatoes



Note. Top to bottom—Day 1 and Day 17; Left to right—Uncoated, Okra, Aloe Vera

Figure 7. Interior of Tomatoes Post-Experimentation



CHAPTER 5: CONCLUSION

The research findings suggest that **okra mucilage cannot be utilized** as a bio-preservative for tomatoes, as there were **no notable differences** observed in the weight and flesh color throughout experimentation across all three groups.

Future studies could explore other parameters such as firmness, texture, microbial activity, and moisture content to better understand the effectiveness of edible coatings in preserving tomatoes.